PROGRAM:

AIES - 1.py - C:/Users/ABBIRAMY/Documents/AIES - 1.py (3.14.2)

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```
rules = [
    ({"power_light_off", "fan_not_spinning"},          "no_power"),
    ({"no_power", "cable_loose"},                      "power_cable_fault"),
    ({"no_power", "cable_ok"},                         "power_supply_fault"),
    ({"screen_black", "fan_spinning", "beep_sound"},   "ram_fault"),
    ({"screen_black", "fan_spinning", "no_beep"},      "display_fault"),
    ({"slow_performance", "high_disk_usage"},          "hard_disk_fault"),
    ({"slow_performance", "high_cpu_temp"},            "overheating"),
    ({"no_internet", "wifi_icon_off"},                 "network_adapter_fault"),
    ({"power_cable_fault"},                            "FIX: Reconnect the power cable"),
    ({"power_supply_fault"},                           "FIX: Replace the power supply unit"),
    ({"ram_fault"},                                    "FIX: Reseat or replace the RAM"),
    ({"display_fault"},                                "FIX: Check monitor cable or graphics card"),
    ({"hard_disk_fault"},                              "FIX: Clean disk or replace hard disk"),
    ({"overheating"},                                  "FIX: Clean the fan and apply thermal paste"),
    ({"network_adapter_fault"},                        "FIX: Reinstall network driver"),
]
facts = {"power_light_off", "fan_not_spinning", "cable_loose"}
def forward_chaining(facts):
    facts = set(facts)
    changed = True
    while changed:
        changed = False
        for conditions, conclusion in rules:
            if conditions <= facts and conclusion not in facts:
                print(f"Rule fired: IF {sorted(conditions)} THEN {conclusion}")
                facts.add(conclusion)
                changed = True
    return facts
print("User facts:", sorted(facts), "\n")
final = forward_chaining(facts)
print("\nConclusions:")
for f in sorted(final - facts):
    print("  *", f)
```

OUTPUT:

```
>>>
    ====================== RESTART: C:/Users/ABBIRAMY/Documents/AIES - 1.py ======================
    User facts: ['cable_loose', 'fan_not_spinning', 'power_light_off']

    Rule fired: IF ['fan_not_spinning', 'power_light_off'] THEN no_power
    Rule fired: IF ['cable_loose', 'no_power'] THEN power_cable_fault
    Rule fired: IF ['power_cable_fault'] THEN FIX: Reconnect the power cable

    Conclusions:
      * FIX: Reconnect the power cable
      * no_power
      * power_cable_fault
>>>
```

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